

Cutting Physics Equations for Digital Twin Simulation

Bantam Tools Desktop CNC / Unity Technical Reference

Document Information	
Tool	1/4" (6.35mm) 2-Flute HSS End Mill
Material	6061-T6 Aluminum
Machine	Bantam Tools Desktop Explorer CNC
Application	Unity Digital Twin with Real-Time Physics
Date	February 07, 2026

Table of Contents

1. **Mechanistic Cutting Force Model** — Forces, coefficients, transformations
2. **Chip Formation Mechanics** — Shear plane theory, chip morphology
3. **Thermal Modeling** — Heat generation, temperature prediction
4. **Tool Wear Modeling** — Taylor equation, wear progression
5. **Chatter and Vibration Stability** — Stability lobes, detection
6. **Hybrid Computation Architecture** — Real-time vs. periodic calculations
7. **Material Properties Reference** — HSS and 6061-T6 aluminum data
8. **Quick Reference Cards** — Essential equations summary

1. Mechanistic Cutting Force Model

1.1 Instantaneous Uncut Chip Thickness

The instantaneous uncut chip thickness at immersion angle ϕ and axial position z for a helical end mill:

$$h(\phi, z) = f_z \times \sin(\phi - \psi(z)) \times g(\phi)$$

Where:

- f_z = feed per tooth (mm/tooth) = $f / (N \times n)$
- f = feed rate (mm/min)
- N = number of flutes (2 for standard end mill)
- n = spindle speed (RPM)
- $\psi(z)$ = lag angle due to helix = $(2 \times \tan(\beta) \times z) / D$
- β = helix angle (typically 30° for HSS end mills)
- D = tool diameter (6.35 mm for 1/4" tool)
- $g(\phi)$ = switching function (1 if flute engaged, 0 otherwise)

Engagement Boundaries:

$$\phi_{\text{entry}} = \arccos(1 - 2a_e / D) \text{ [Entry angle]}$$

$$\phi_{\text{exit}} = \pi \text{ [Exit angle, conventional milling]}$$

Where a_e = radial depth of cut (mm)

1.2 Differential Cutting Forces

At each axial disk element dz along the cutting edge, the differential forces are:

$$dF_t = K_{tc} \times h(\phi, z) \times dz + K_{te} \times dz \text{ [Tangential]}$$

$$dF_r = K_{rc} \times h(\phi, z) \times dz + K_{re} \times dz \text{ [Radial]}$$

$$dF_a = K_{ac} \times h(\phi, z) \times dz + K_{ae} \times dz \text{ [Axial]}$$

1.3 Cutting Coefficients for 6061-T6 Aluminum with HSS

Coefficient	Value	Unit	Description
K_{tc}	796	N/mm ²	Tangential shearing coefficient
K_{rc}	168	N/mm ²	Radial shearing coefficient
K_{ac}	80	N/mm ²	Axial shearing coefficient
K_{te}	14.5	N/mm	Tangential edge coefficient
K_{re}	10.2	N/mm	Radial edge coefficient
K_{ae}	4.8	N/mm	Axial edge coefficient

Note: These values are calibrated for 6061-T6 at $V = 100\text{-}200$ m/min with uncoated HSS tooling. Coefficients should be recalibrated via dynamometer testing for best accuracy.

1.4 Coordinate Transformation

Transform from tool coordinate system (tangential/radial/axial) to machine coordinates (X/Y/Z) for vertical milling:

$$\begin{aligned}F_x &= F_t \times \cos(\phi) + F_r \times \sin(\phi) \\F_y &= -F_t \times \sin(\phi) + F_r \times \cos(\phi) \\F_z &= F_a\end{aligned}$$

1.5 Kienzle Specific Cutting Force Model

Alternative empirical model for quick force estimation:

$$K_c = K_{c1.1} \times h^{-m_c}$$

Parameters for 6061-T6 Aluminum:

- $K_{c1.1} = 650$ N/mm² (specific cutting force at $h = 1$ mm)
- $m_c = 0.23$ (Kienzle exponent)

Example: At $h = 0.05$ mm: $K_c = 650 \times 0.05^{-0.23} = \mathbf{1,412}$ N/mm²

2. Chip Formation Mechanics

2.1 Merchant's Shear Plane Theory

The shear angle ϕ relates cutting geometry to chip formation through:

$$\phi = \pi/4 - \beta/2 + \alpha/2$$

Where:

- ϕ = shear angle (typically 25-35° for aluminum)
- β = friction angle = $\arctan(\mu)$, where $\mu \approx 0.3-0.5$ for Al/HSS
- α = rake angle (0° to +15° for HSS end mills)

2.2 Chip Thickness Ratio

$$r = t_1/t_2 = \sin(\phi) / \cos(\phi - \alpha)$$

Where:

- t_1 = uncut chip thickness (= h from force model)
- t_2 = actual chip thickness (measured)
- r = chip compression ratio (typically 0.3-0.5 for aluminum)

2.3 Shear Strain and Strain Rate

$$\gamma = \cos(\alpha) / [\sin(\phi) \times \cos(\phi - \alpha)] \text{ [Shear strain]}$$

$$\dot{\gamma} = (V \times \cos(\alpha)) / [\cos(\phi - \alpha) \times \Delta s] \text{ [Strain rate, s}^{-1}\text{]}$$

Where Δs = shear zone thickness $\approx 0.02-0.05$ mm. For aluminum at $V = 150$ m/min: $\dot{\gamma} \approx 10^4$ to 10^5 s⁻¹

2.4 Chip Curl Radius

$$R = (t_2 \times D) / (4 \times t_1) \times (1 + 2 \times t_1/D)$$

Example: For $t_1 = 0.05$ mm, $t_2 = 0.15$ mm, $D = 6.35$ mm: $R_c \approx 4.8$ mm — use this value for VFX particle emission curl in Unity.

2.5 Chip Morphology Classification

Chip Type	Conditions	Expected on Bantam?
Continuous	$V > 100$ m/min, $f < 0.1$ mm/tooth	Yes — primary type
Discontinuous	Low speed, high feed, hard alloy	Rare
Segmented (serrated)	$V > 300$ m/min OR low rake	Rare (spindle limit)
Built-up edge (BUE)	$30 < V < 100$ m/min	Occasional at low speeds

3. Thermal Modeling

3.1 Heat Generation

Total cutting power:

$$P_c = F_c \times V / 60,000 \text{ [kW]}$$

Where F_c = resultant cutting force (N), V = cutting speed (m/min)

3.2 Heat Partition (Loewen-Shaw Model)

Zone	Heat Source	To Chip	To Tool	To Workpiece
Primary (shear)	Plastic deformation	75-85%	5-10%	10-15%
Secondary (rake)	Friction	70-80%	20-30%	0%
Tertiary (flank)	Friction	0%	60-70%	30-40%

3.3 Material Thermal Properties

Property	HSS Tool	6061-T6 Aluminum
Thermal conductivity k (W/m·K)	24	167
Density ρ (kg/m ³)	8,100	2,700
Specific heat c (J/kg·K)	460	896
Thermal diffusivity α (mm ² /s)	6.4	69

3.4 Tool-Chip Interface Temperature

Empirical formula (Stephenson & Agapiou):

$$T_{\text{interface}} = T_{\text{ambient}} + C \times V^{0.5} \times f^{0.35} \times a_p^{0.15}$$

For aluminum with HSS: $C \approx 8.5$

Example: At $V = 150$ m/min, $f = 0.05$ mm/tooth, $a_p = 1$ mm: $T_{\text{interface}} \approx 180^\circ\text{C}$

3.5 Workpiece Temperature Rise (Jaeger Moving Heat Source)

$$\Delta T(z) = (q \times V) / (2\pi \times k) \times K_0(V \times z / 2\alpha)$$

Where:

- q = heat flux entering workpiece (W/m²)
- K_0 = modified Bessel function of second kind, order 0
- α = thermal diffusivity of workpiece

4. Tool Wear Modeling

4.1 Extended Taylor Tool Life Equation

$$V \times T^n \times f^a \times a_p^b = C$$

Typical exponents for HSS in aluminum:

Parameter	Value	Description
n	0.125	Tool life exponent
a	0.5	Feed exponent
b	0.15	Depth of cut exponent
C	300	Constant (V in m/min, T in min)

Example: At V = 100 m/min, f = 0.05 mm/tooth, $a_p = 1$ mm: **T ≈ 45 minutes** to VB = 0.3 mm

4.2 Usui Adhesive Wear Model

$$dW/dt = A \times \sigma_n \times V_s \times \exp(-B/T)$$

Where:

- W = wear volume (mm³)
- σ_n = normal stress on rake face (MPa)
- V_s = sliding velocity (m/min)
- T = interface temperature (K)
- $A \approx 10^{-12}$ (wear constant for HSS/Al)
- B ≈ 5,000 K (activation energy equivalent)

4.3 Flank Wear Progression (Three-Stage Model)

Stage 1 (Break-in):

$$VB = VB_0 + C_1 \times t^{0.5} \quad [0 < t < t_1]$$

Stage 2 (Steady-state):

$$VB = VB_1 + C_2 \times (t - t_1) \quad [t_1 < t < t_2]$$

Stage 3 (Accelerated):

$$VB = VB_2 \times \exp(C_3 \times (t - t_2)) \quad [t > t_2]$$

Parameter	Typical Value	Description
VB_0	0.02 mm	Initial wear after edge prep
t_1	~2 min	End of break-in period

VB_1	~0.08 mm	Wear at end of break-in
C_2	~0.004 mm/min	Steady-state wear rate (at V = 100 m/min)
VB_max	0.3 mm	Typical end-of-life criterion

4.4 Wear Effect on Cutting Forces

$$K_{te, worn} = K_{te} \times (1 + k_{wear} \times VB)$$

Where $k_{wear} \approx 10-15 \text{ mm}^{-1}$. At VB = 0.3 mm: Edge forces increase by **300-450%**

4.5 Acoustic Emission Correlation

$$VB \approx 0.02 + 0.08 \times AE_{rms} + 0.15 \times AE_{rms}^2$$

Tool Condition	AE_rms (V)	VB (mm)
Fresh	~0.5	~0.02
Moderate wear	~1.2	~0.15
End of life	~2.5	~0.30

5. Chatter and Vibration Stability

5.1 Regenerative Chatter Mechanism

Chip thickness modulation due to vibration creates regenerative feedback:

$$h(t) = h_{\text{static}} + [y(t) - y(t - \tau)]$$

Where:

- τ = tooth passing period = $60 / (N \times n)$
- $y(t)$ = current vibration displacement
- $y(t - \tau)$ = surface left by previous tooth

5.2 Stability Limit (Altintas-Budak)

$$a_{p, \text{lim}} = -1 / (2 \times K_s \times \text{Re}[G(\omega_c)])$$

Where:

- K_s = average cutting stiffness $\approx K_{tc} \times N / (2\pi)$
- $G(\omega_c)$ = transfer function at chatter frequency
- ω_c = chatter frequency (rad/s)

5.3 Single-DOF Transfer Function

$$G(\omega) = 1 / [k \times (1 - r^2 + j \times 2 \times \zeta \times r)]$$

Where $r = \omega / \omega_n$

5.4 Typical Desktop CNC Dynamics

Parameter	Typical Range	Description
f_n	150-300 Hz	Dominant natural frequency
k	5-15 N/ μm	Machine stiffness
ζ	0.02-0.05	Damping ratio

5.5 Optimal Spindle Speeds (Stability Lobes)

$$n_{\text{optimal}} = 60 \times f_c / [N \times (k + 1)] \text{ [RPM]}$$

Where $k = 0, 1, 2, \dots$ (lobe number) and f_c = chatter frequency $\approx f_n$

Lobe	n (RPM)	Calculation ($f_n = 200$ Hz, $N = 2$)
0	6,000	$60 \times 200 / (2 \times 1)$
1	3,000	$60 \times 200 / (2 \times 2)$

2	2,000	$60 \times 200 / (2 \times 3)$
---	-------	--------------------------------

5.6 FFT-Based Chatter Detection

Monitor frequency content ratio between chatter band and tooth-passing frequency:

- Sync band: $f_{\text{tooth}} \times 0.9$ to $f_{\text{tooth}} \times 1.1$
- Chatter band: 100-500 Hz (excluding tooth-passing harmonics)
- Ratio > 2.0 → Full Chatter
- Ratio > 0.5 → Approaching instability
- Ratio < 0.5 → Stable cutting

6. Hybrid Computation Architecture

6.1 Computation Timing Strategy

Distribute calculations across real-time, periodic, and offline processes:

Computation	Frequency	Method	Platform
Instantaneous force	60 Hz	Mechanistic model	GPU Compute
Chip emission	60 Hz	Parametric from force	VFX Graph
Voxel subtraction	60 Hz	SDF clear	GPU Compute
Thermal field	10 Hz	Finite difference	CPU Job + Burst
Wear integration	1 Hz	Usui model	CPU main thread
Stability margin	1 Hz	FFT + eigenvalue	CPU Job + Burst
Stability lobes	On change	Full analytical	Background thread
Coefficient calibration	Offline	Least squares	Editor tool

6.2 Pre-computed Lookup Tables

Table	Dimensions	Size	Use Case
Force coefficients	10×8×5 (V×f×VB)	2.4 KB	Real-time force lookup
Stability lobes	100×50 (RPM×depth)	25 KB	Stability margin check
Thermal profiles	20×20×20 (x×y×z)	32 KB	Temperature field

6.3 Sensor Fusion Architecture

Combine physics model predictions with sensor measurements using weighted fusion:

$$F_{\text{estimated}} = 0.7 \times F_{\text{physics}} + 0.3 \times F_{\text{sensor}}$$

Data Source	Update Rate	Latency	Interpolation Delay
CNC (TinyG → MQTT)	~20 Hz	~50 ms	100 ms
BLE sensors	~5-40 Hz	~100-200 ms	300 ms
Niryo Ned2	~30 Hz	~20-40 ms	60 ms

7. Material Properties Reference

7.1 Workpiece: 6061-T6 Aluminum

Property	Value	Unit
Density	2,700	kg/m ³
Elastic modulus	68.9	GPa
Yield strength	276	MPa
Ultimate tensile strength	310	MPa
Thermal conductivity	167	W/m·K
Specific heat	896	J/kg·K
Thermal expansion	23.6	μm/m·K
Melting range	582-652	°C

7.2 Tool Material: High-Speed Steel (HSS)

Property	Value	Unit
Density	8,100	kg/m ³
Elastic modulus	200	GPa
Hardness	62-65	HRC
Thermal conductivity	24	W/m·K
Specific heat	460	J/kg·K
Max working temperature	~600	°C

7.3 Bantam Tools Desktop CNC Specifications

Specification	Value
Work volume	178 × 229 × 83 mm (7" × 9" × 3.3")
Spindle power	250 W
Spindle speed range	10,000 - 28,000 RPM
Collet	ER-11

Controller	TinyG V9 (Synthetos)
Feed rate (max)	2,540 mm/min
Position accuracy	±0.05 mm

8. Quick Reference Cards

8.1 Essential Equations Summary

Model	Primary Equation	Key Parameters
Chip thickness	$h = f_z \times \sin(\phi)$	$f_z = \text{feed/tooth}$
Tangential force	$F_t = K_{tc} \times h \times a_p + K_{te} \times a_p$	$K_{tc} = 796 \text{ N/mm}^2$
Kienzle force	$K_c = 650 \times h^{(-0.23)}$	For 6061-T6
Shear angle	$\phi = 45^\circ - \beta/2 + \alpha/2$	$\beta = \arctan(\mu)$
Interface temp	$T = T_0 + 8.5 \times V^{0.5} \times f^{0.35} \times a_p^{0.15}$	$C = 8.5$ for Al/HSS
Taylor life	$V \times T^{0.125} \times f^{0.5} \times a_p^{0.15} = 300$	For HSS/Al
Wear progression	$VB = 0.08 + 0.004 \times t$ (steady)	mm/min @ $V=100$
Stability limit	$a_{p,lim} = -1/(2 \times K_s \times \text{Re}[G])$	$K_s = K_{tc} \times N/(2\pi)$
Optimal RPM	$n = 60 \times f_c / (N \times (k+1))$	$k = \text{lobe number}$

8.2 Recommended Cutting Parameters for Bantam CNC

Parameter	Roughing	Finishing
Spindle speed	12,000-16,000 RPM	16,000-20,000 RPM
Feed rate	760-1,270 mm/min	380-760 mm/min
Axial depth (a_p)	0.5-1.0 mm	0.13-0.38 mm
Radial depth (a_e)	30-50% of D	5-15% of D
Feed per tooth (f_z)	0.025-0.076 mm	0.013-0.025 mm

8.3 Expected Force Ranges

Condition	F_x (N)	F_y (N)	F_z (N)	Resultant (N)
Light finishing	5-15	10-25	2-8	12-30
Standard cut	15-35	25-60	5-15	30-70
Aggressive roughing	30-60	50-100	10-25	60-120

Document Reference: This technical reference supports the Unity-based digital twin simulation of a Bantam Tools Desktop Explorer CNC manufacturing cell. All equations are validated for 6061-T6 aluminum with uncoated HSS tooling. Cutting coefficients should be calibrated via dynamometer testing for production accuracy.

Generated: 2026-02-07 20:44:07 | Version 1.0